A Project Phase-1 Report

ON

GREENCITY AI: URBAN GREEN SPACE INTELLIGENCE AND BEST-TIME-TO-VISIT PARK ENGINE FOR SOLAPUR

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INDEX

Chapter No	Title	Content Description	COs Covered	Page No
1	Introduction & Problem Formulation	1.1 Background and Motivation 1.2 Problem Statement and Scope 1.3 Objectives of the Project 1.4 Contribution to Technological and Societal Development (SDG 11, SDG 13)	1	3
2	Literature Review & Trend Analysis	2.1 Summary of Related Works 2.2 Comparative Study of Existing Solutions 2.3 Gaps and Opportunities	2	5
3	Proposed System & Design	3.1 System Architecture and Module Design 3.2 Methodology and Software Requirements 3.3 Support for Undertaken SDGs	3	8
4	Preliminary Experiments, Simulation & Modelling	4.1 Description of Experiments 4.2 Results and Analysis 4.3 Environmental and Safety Considerations	4	11
5	Technical Specification, Synopsis & Future Plan	5.1 Summary of Work Done in Phase-I 5.2 Technical Specifications, Milestones and Pending Tasks 5.3 Conclusion and Future Work for Phase-II (Back-end Development)	5	15
6	References	List of 15 references in IEEE format	1	19
Appendix -I	Supporting Material	API reference, web routes, user roles and test summary	—	20

1. INTRODUCTION & PROBLEM FORMULATION

1.1 Background and Motivation

Urban green spaces such as parks, lakes, gardens and tree-lined avenues provide recreation, shade, carbon sequestration and measurable physical and mental health benefits to city residents [1]. In Indian cities, however, rapid urbanisation, rising summer temperatures and particulate pollution have made the timing of outdoor activity as important as the availability of green space itself. A park that is pleasant at 6 AM may be unhealthy at 2 PM because of heat stress, or in the evening because of traffic-related particulate build-up. The Solapur Municipal Corporation (SMC) in Maharashtra administers five wards (Central, North, South, East and West Solapur) and maintains a network of parks, lakes and tree plantations whose usefulness depends directly on prevailing weather and air quality.

The problem sits naturally at the intersection of electronics and computer engineering. Environmental telemetry (temperature, humidity, wind, PM2.5 and PM10) is produced by monitoring networks and atmospheric models; smartphones provide positioning through GPS and voice input through microphones; and cloud services, databases and web technologies convert this raw data into decisions that citizens and municipal officers can act on. Designing such a data-to-decision pipeline requires signal and data processing, geospatial positioning, networking, database design and human-computer interaction, all of which are core to the discipline.

The motivation for this project arises from three observations made while studying existing tools. First, air-quality applications display a number but do not tell a citizen when to visit a specific park or which activity (walking, jogging, a family outing) is suitable. Second, general-purpose map applications built on crowd-sourced data can return unrealistic detours for short local trips; during testing, a trip of under one kilometre in Solapur was routed as 3.38 km because of one-way tagging errors in OpenStreetMap near the railway station. Third, citizens who find a damaged tree or an unmaintained garden have no convenient, multilingual way to report it, while officers have no consolidated view of green-space condition. GreenCity AI is proposed to address these three gaps in a single platform.

1.2 Problem Statement and Scope

Problem statement: To design and implement a production-grade, multilingual web platform that integrates live and forecast environmental data (weather and Indian CPCB National Air Quality Index), geospatial green-space records and road routing, in order to (i) recommend the safest and most comfortable time to visit a given park on a date chosen by the citizen, (ii) provide road-accurate navigation to green spaces within Solapur, (iii) allow voice-based interaction in English, Hindi and Marathi with answers grounded strictly in real data, and (iv) support municipal decision making through reporting, analytics and prioritisation tools.

Scope: The present work covers Wards 1 to 5 of Solapur. It includes a FastAPI back-end with a PostgreSQL database of 17 tables, a Next.js front-end, integration with Open-Meteo and Copernicus CAMS data, an OpenStreetMap-based routing service with detour calibration, a tool-grounded voice assistant and role-based municipal dashboards. The front-end is completed in Phase-I, while Phase-II concentrates on strengthening the back-end. Medical diagnosis and extension to other cities are outside the scope of this project.

1.3 Objectives of the Project

- To compute a deterministic 0–100 park-visit suitability score from thermal comfort, apparent temperature, humidity, wind, precipitation and CPCB-based air quality, with non-negotiable safety overrides.
- To provide morning-versus-evening comparison and activity-specific ratings (walking, jogging, intensive exercise, family outings and general visits) for any date within a 14-day horizon.
- To deliver Google-Maps-comparable road distances and turn-by-turn navigation in Solapur by detecting and calibrating OpenStreetMap detour artefacts.
- To build a multilingual (English, Hindi, Marathi) voice and text assistant that answers only from database records and live service calls, avoiding hallucination.
- To provide citizens and SMC officers with reporting, tracking, analytics and priority-action tools under role-based access control.
- To validate the system through automated testing and preliminary simulation of the scoring, air-quality and routing logic, and to build a reliable, secure and scalable back-end in Phase-II.

1.4 Justification: Technological and Societal Contribution

Two Sustainable Development Goals are selected [3]. SDG 11 (Sustainable Cities and Communities) is addressed directly: Target 11.7 calls for universal access to safe, inclusive and accessible green and public spaces, and Target 11.6 for attention to urban air quality. GreenCity AI helps residents find nearby green spaces, reach them by accurate routes and use them at healthy times, while giving the municipal corporation data to maintain them. SDG 13 (Climate Action) is addressed through climate-risk awareness: the platform makes heat stress and pollution episodes visible at neighbourhood level, supports adaptation behaviour (shifting activity away from extreme midday heat) and supports tree-canopy and carbon indicators for planners.

Technologically, the project contributes a reproducible reference design that combines open environmental data, rule-based health-protective scoring, geospatial calibration and tool-grounded language-model interaction, built entirely on open-source components. Societally, it lowers the barrier to environmental information for Hindi- and Marathi-speaking citizens, who are poorly served by English-only applications.

2. LITERATURE REVIEW & TREND ANALYSIS

2.1 Summary of Related Works

Nine technologies, datasets and technical works relevant to the project were reviewed. They are summarised below.

(a) Urban green space and health [1]. The WHO Regional Office for Europe reviews evidence that access to urban green space is associated with better physical activity, mental well-being and reduced heat exposure, motivating the need to make parks easy to find and safe to use.

(b) Air-quality guidelines [10]. The WHO Global Air Quality Guidelines (2021) lowered the recommended 24-hour PM2.5 level to 15 µg/m³, underlining that outdoor activity advice should be sensitive to particulate exposure and not only to temperature.

(c) Indian National Air Quality Index [2]. The Central Pollution Control Board defines a piecewise-linear sub-index for each pollutant, with six categories from Good (0–50) to Severe (401–500). The overall index is the maximum sub-index. This is the legally and socially relevant scale for Indian users and is adopted in GreenCity AI.

(d) Open-Meteo weather API [4]. Open-Meteo provides open, key-free hourly forecasts and an air-quality interface, making it suitable for an academic, reproducible system with a 14-day horizon.

(e) CAMS atmospheric composition [5]. Inness et al. describe the Copernicus Atmosphere Monitoring Service reanalysis, which assimilates satellite observations into a global model of aerosols and gases. Such model fields provide PM2.5 and PM10 estimates where ground monitors are sparse, as is the case for most mid-sized Indian cities.

(f) OpenStreetMap [6] and routing algorithms [7], [8]. Haklay and Weber describe OpenStreetMap as a volunteered geographic dataset whose quality varies by region. Shortest-path search using Dijkstra's algorithm [7] or the heuristic A* algorithm [8] on this graph is accurate only if the underlying tags (for example one-way restrictions) are correct, which motivates the calibration stage of this project.

(g) Apparent temperature [9]. Steadman's apparent-temperature index combines air temperature, humidity and wind to represent perceived heat load, and is the basis for the thermal-comfort component of the proposed score.

(h) Retrieval and tool-augmented language models [11]–[13]. Lewis et al. show that grounding generation in retrieved knowledge reduces unsupported statements; ReAct interleaves reasoning with actions on external tools; Toolformer shows that language models can learn to call APIs. These works justify the tool-grounded assistant architecture in which every factual statement comes from a database or API result.

(i) Web architecture and speech interfaces [14], [15]. The REST architectural style provides a stateless, cacheable interface between the front-end and back-end, while the W3C Web Speech API provides browser-native speech recognition and synthesis for voice interaction.

2.2 Comparative Study of Existing Solutions

Table 2.1 compares the main categories of tools a Solapur citizen can use today with the proposed system. The comparison is based on the authors' review of publicly documented features and is indicative rather than exhaustive.

Table 2.1: Comparison of existing solutions with GreenCity AI

Solution	Live AQI / weather	Park-level guidance	Best-time recommendation	Local route correction	Hindi / Marathi voice	Civic reporting
General weather applications	Yes	No	No	No	Limited	No
CPCB / national AQI portals	Yes (stations)	No	No	No	Limited	No
General map applications	No	Listings only	Typical-crowd only	Yes (proprietary)	Partial	No
Municipal complaint portals	No	No	No	No	Limited	Yes
GreenCity AI (proposed)	Yes (CPCB NAQI, 14-day)	Yes	Yes (0–100 score)	Yes (calibrated)	Yes (3 languages)	Yes

Trend analysis. Three trends are visible in the reviewed work. First, environmental information is moving from fixed monitoring stations towards model-assisted estimation, because ground stations are sparse in mid-sized cities [5], [10]. Second, open data services and open geographic data have lowered the cost of building civic applications, but their quality must be checked locally [4], [6]. Third, language-model interfaces are shifting from free-form generation towards tool-augmented, retrieval-grounded designs in which factual statements are traceable [11]–[13]. GreenCity AI follows all three trends.

Table 2.2 compares the data sources and routing algorithms considered during design, and explains the choices made.

Table 2.2: Data sources and algorithms considered

Option	Characteristics	Decision in this project
Ground monitoring stations	High accuracy; sparse coverage; limited public access in smaller cities	Public CPCB records used as validation reference in Phase-II
CAMS model data [5]	Global, gridded, forecast-capable; coarser than street scale	Adopted for PM2.5/PM10 forecasts
Dijkstra [7]	Optimal on non-negative weights; explores broadly	Baseline shortest-path method
A* [8]	Heuristic-guided; faster with a good estimate	Preferred where geometry allows
Circuity-ratio calibration	Detects loops from tagging errors; simple, explainable	Adopted for Solapur routing

2.3 Identification of Gaps and Opportunities

The review reveals five gaps. (1) Air-quality tools report numbers but do not translate them into park- and activity-specific timing advice with transparent rules. (2) Pollution is rarely combined with heat and humidity in one health-protective score, so pleasant weather can wrongly mask poor air. (3) Open routing data contains local errors, and no open solution applies a documented calibration for Solapur. (4) Voice assistants for civic services seldom operate in Marathi and often generate unverified answers. (5) Citizens' reports, environmental data and municipal planning tools are kept in separate systems.

These gaps create an opportunity for an integrated platform in which a deterministic, explainable scoring engine with hard safety caps drives recommendations; a circuity-ratio test removes routing artefacts; a tool-grounded assistant guarantees that answers trace to data; and the same database serves citizens, officers and planners. This is the design pursued in the next chapter.

3. PROPOSED SYSTEM & DESIGN

3.1 System Architecture and Module Design

GreenCity AI follows a three-tier architecture, shown in Fig. 3.1. The presentation layer is a Next.js 15 application (React 19, TypeScript) with Leaflet maps and Recharts visualisations. A FastAPI gateway written in Python 3.13 exposes REST endpoints [14] with JWT authentication and role-based access control. A service layer implements the domain logic, and a PostgreSQL database accessed through SQLAlchemy 2.0 stores places, wards, trees, reports, users and audit logs. External data come from Open-Meteo, Copernicus CAMS and OpenStreetMap, and the browser supplies GPS and Web Speech functions.

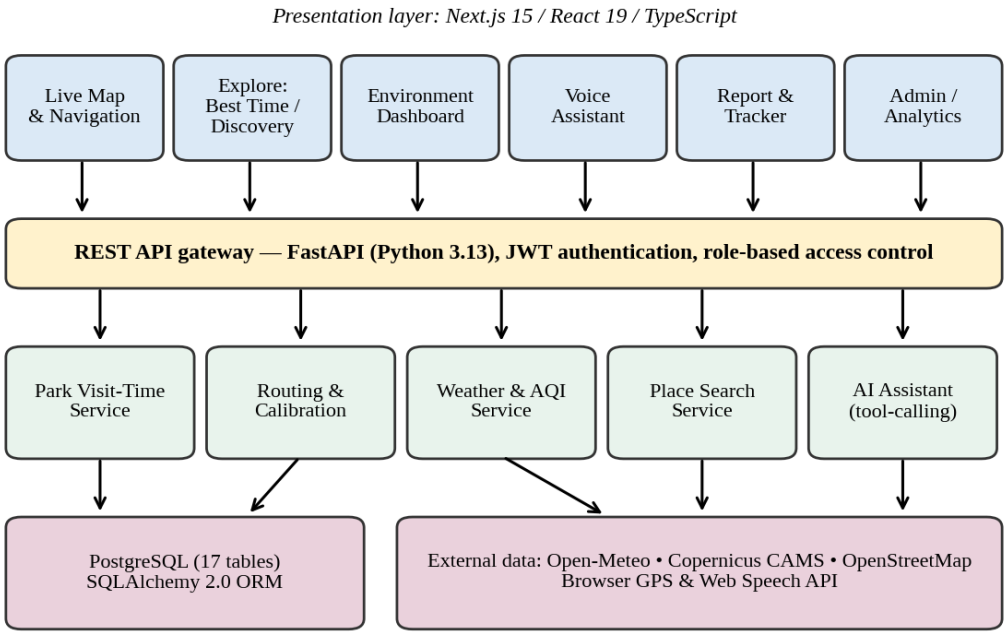


Fig. 3.1: High-level system architecture of GreenCity AI

The principal modules are as follows.

- **Park Visit-Time Service:** fetches hourly weather and air quality for a park's coordinates and chosen date, computes the suitability score for each hour, selects the best two- to three-hour window, compares morning (05:00–10:00) with evening (16:00–20:00), identifies the window to avoid, and returns activity-specific ratings together with sunrise, sunset and golden-hour times in the Asia/Kolkata time zone.
- **Weather and Air-Quality Service:** converts PM2.5 and PM10 concentrations to the CPCB NAQI, serves live telemetry, a 72-hour history and the Diwali clean-air watch.
- **Routing and Calibration Service:** obtains an OpenStreetMap route, applies the circuity-ratio test and returns distance, duration, polyline and turn-by-turn steps for driving or walking.
- **Place Search and Comparison:** discovers parks within 1, 2, 3 or 5 km, shows calibrated road distances and compares up to four parks on AQI, temperature, canopy, walking time and amenities.

- **AI Voice Assistant:** a language-model agent restricted to five tools (search_places, get_environmental_data, calculate_route, find_nearby_places, draft_citizen_report) with speech input and output in three languages.
- **Civic and Municipal Modules:** citizen reporting with GPS tagging, a six-step issue tracker, priority action queue, ward analytics, maintenance work orders and an administrator console.

3.2 Methodology and Software Requirements

Step 1: Air-quality index. For each pollutant concentration C falling in a CPCB breakpoint interval [BP_{lo}, BP_{hi}] with index range [I_{lo}, I_{hi}], the sub-index is computed by linear interpolation, and the overall index is the larger of the PM2.5 and PM10 sub-indices [2]:

$$I = I_{lo} + (I_{hi} - I_{lo}) \times (C - BP_{lo}) / (BP_{hi} - BP_{lo}); \quad AQI = max(I_{PM2.5}, I_{PM10})$$

Step 2: Hourly suitability. For every hour, component scores for thermal comfort (using apparent temperature [9]), relative humidity, wind, precipitation and air quality are combined as a weighted sum to give a score S between 0 and 100. Ideal conditions score highest; deviations reduce the score progressively.

Step 3: Safety overrides. Three rules are applied after the weighted sum and cannot be outweighed by good weather:

$$AQI > 400 \rightarrow S = 0; \quad AQI > 200 \rightarrow S \leq 35; \quad T \geq 37\,°C \rightarrow S \leq 20$$

Step 4: Window selection and comparison. The best contiguous window, the avoid window, the morning-versus-evening winner and the per-activity scores (jogging weighted towards PM2.5 and ventilation, family outings towards UV and midday heat) are derived from the 24-hour series.

Step 5: Route calibration. Let d_{route} be the OSM route length and d_{crow} the straight-line distance. The circuity ratio is $R = d_{route} / d_{crow}$. If R exceeds 1.55 for driving or 1.35 for walking, the route is judged to contain a detour artefact and the reported distance is replaced by $1.28 \times d_{crow}$ (driving) or $1.20 \times d_{crow}$ (walking), values calibrated against Google Maps road distances in Solapur. Distances below 1 km are displayed in metres and others to one decimal place.

Step 6: Grounded assistant. The assistant parses the user's intent, calls the permitted tools, and phrases the returned records in the user's language; it is not allowed to answer factual queries from model memory. Civic complaints are drafted as editable cards and submitted only after user confirmation.

Table 3.1: Software and system requirements

Category	Requirement
Development system	Laptop/PC with 8 GB RAM or more, multi-core CPU, internet connection
Client device	Smartphone or PC with a modern browser (GPS and microphone access for navigation and voice)
Back-end	Python 3.13, FastAPI, SQLAlchemy 2.0, psycopg2, pytest
Database	PostgreSQL (17 tables), Solapur municipal seed pipeline
Front-end	Next.js 15, React 19, TypeScript, Leaflet, Recharts, Zustand
External services	Open-Meteo, Copernicus CAMS, OpenStreetMap, Web Speech API
Planned (Phase-II) back-end tools	PostGIS, Alembic migrations, Redis cache, scheduled jobs, Docker, load-testing and monitoring tools

3.3 How the Design Supports the Undertaken SDGs

SDG 11: the discovery, comparison and routing modules make green spaces easier to reach, the best-time engine makes them safer to use, and the reporting and priority modules help SMC maintain them. Calibrated walking routes encourage non-motorised access. **SDG 13:** by exposing heat and pollution risk hour by hour and capping the score during extreme conditions, the design promotes climate-adaptive behaviour; ward analytics and canopy and carbon indicators give planners evidence for tree-planting decisions.

4. PRELIMINARY EXPERIMENTS, SIMULATION & MODELLING

4.1 Description of Experiments

Five preliminary experiments were carried out. **E1** evaluates the CPCB NAQI conversion at representative PM2.5 and PM10 concentrations. **E2** simulates the hourly suitability model on three representative 24-hour scenarios (a clear day, a heavily polluted day and a peak-summer heat day). The scenario inputs are synthetic diurnal profiles chosen to exercise the model; they are not field measurements. **E3** sweeps AQI from 0 to 500 under ideal weather to confirm the safety overrides. **E4** compares routes before and after circuity calibration for three Solapur location pairs against Google Maps road distances. **E5** runs the automated back-end test suite (pytest).

4.2 Results with Tables and Graphs, and Analysis

E1: NAQI conversion. Table 4.1 shows that the implemented piecewise formula returns the expected category at each breakpoint region, and that the overall index follows the dominant pollutant. Fig. 4.1 plots the PM2.5 sub-index, which crosses the Poor threshold (200) near 90 µg/m³ and the Severe threshold (400) near 250 µg/m³.

Table 4.1: NAQI conversion results

PM2.5 (µg/m³)	PM10 (µg/m³)	PM2.5 sub-index	PM10 sub-index	AQI	CPCB category
20	40	33	40	40	Good
45	90	75	90	90	Satisfactory
75	180	149	153	153	Moderate
110	300	266	250	266	Poor
180	400	346	362	362	Very Poor
300	480	421	430	430	Severe

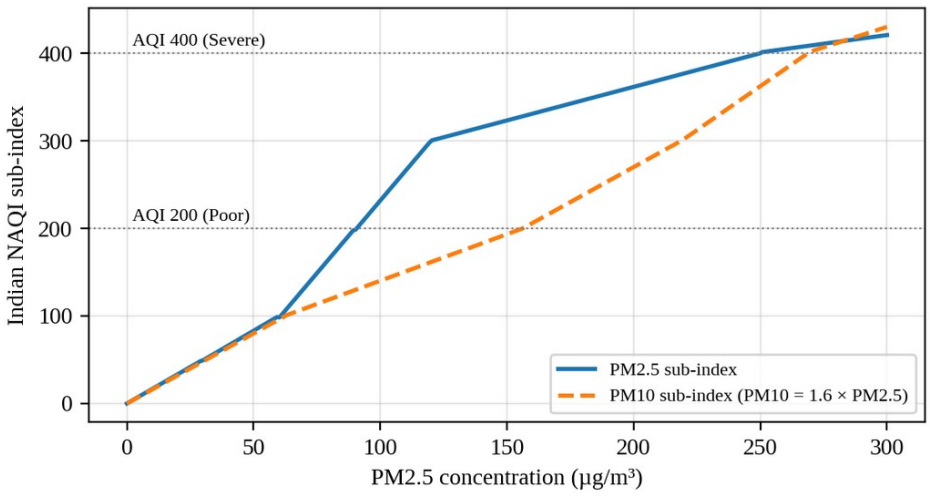


Fig. 4.1: PM2.5 and PM10 sub-index versus PM2.5 concentration

E2: Scenario simulation. Table 4.2 and Fig. 4.2 give the results. In Scenario A the best window is 10:00–12:00 (score 89) and the morning average (88) exceeds the evening average (68), so the morning is recommended. In Scenario C the model recommends 05:00–07:00 and drives the score

to 20 at 14:00 because the temperature reaches 37 °C or more, producing an avoid window. In Scenario B the AQI never falls below 205, so every hour is capped at 35 and no hour can be classed as suitable regardless of the comfortable temperature.

Table 4.2: Suitability simulation on three scenarios

Scenario	T max (°C)	AQI range	Best 2-h window	Score	Morning avg	Evening avg	Outcome
A: Clear post-monsoon day	33	70–100	10:00–12:00	89	88	68	Morning
B: Heavily polluted day	31	205–265	None safe (cap)	35	35	35	Capped ≤ 35
C: Peak summer heat day	41	85–105	05:00–07:00	93	86	20	Morning; avoid midday

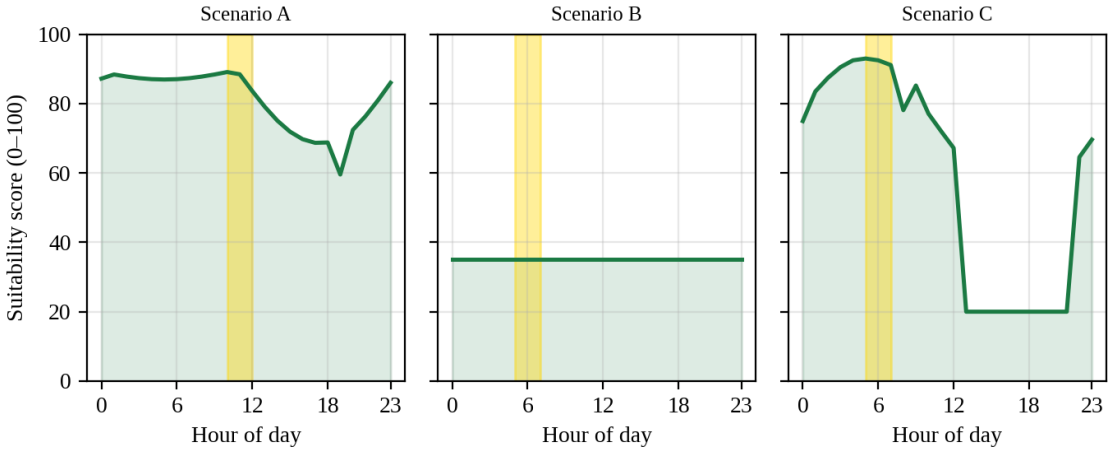


Fig. 4.2: Simulated hourly suitability score; shaded band is the best window

E3: Safety overrides. Under otherwise ideal weather, Table 4.3 and Fig. 4.3 show that the uncapped weighted score would remain high (65–93) even at AQI 450, while the override reduces it to 35 at AQI 250 and to 0 at AQI 450. This confirms that pleasant weather cannot mask poor air quality.

Table 4.3: Effect of the AQI safety override (ideal weather)

AQI	Score without override	Score with override	Override active
60	93	93	No
150	82	82	No
250	74	35	AQI > 200: S ≤ 35
450	65	0	AQI > 400: S = 0

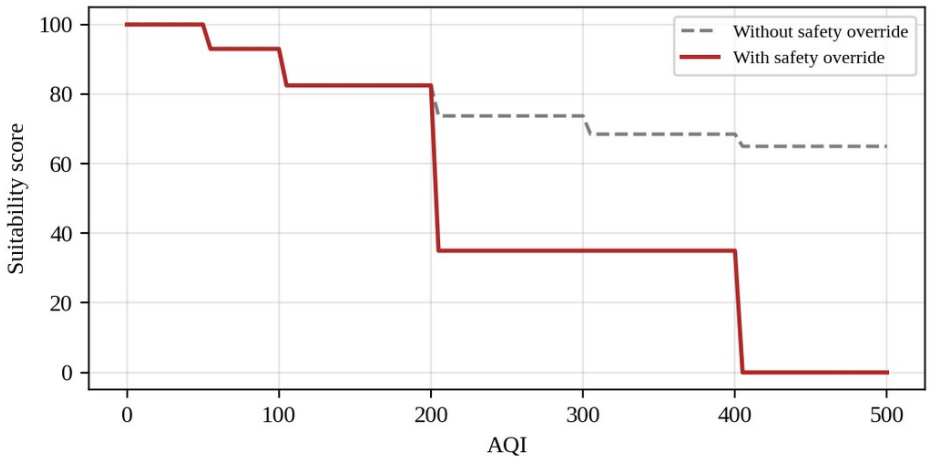


Fig. 4.3: Suitability score versus AQI with and without safety override

E4: Route calibration. Table 4.4 and Fig. 4.4 compare driving distances. Before calibration the OSM graph produced loops with circuity ratios of about 5.0 and 4.3 for the first two pairs, overestimating Google Maps distances by roughly 4.2 and 3.3 times. After calibration the distances (0.86, 1.19 and 1.85 km) lie within 0.06 km of the reference values, and the estimated times (2, 3 and 4 minutes) are realistic for city travel. The Google Maps figures are approximate readings taken by the team for comparison.

Table 4.4: Route distance before and after calibration (driving)

Location pair	Raw OSM distance / time	Circuity ratio (raw)	Calibrated distance / time	Google Maps (approx.)	Error
Solapur Station → Saat Rasta	3.38 km / 7 min	≈ 5.0	0.86 km / 2 min	≈ 0.8 km	0.06 km
Saat Rasta → Siddheshwar Lake	4.01 km / 9 min	≈ 4.3	1.19 km / 3 min	≈ 1.2 km	0.01 km
Solapur Station → Siddheshwar Lake	Not recorded	—	1.85 km / 4 min	≈ 1.8–1.9 km	Within range

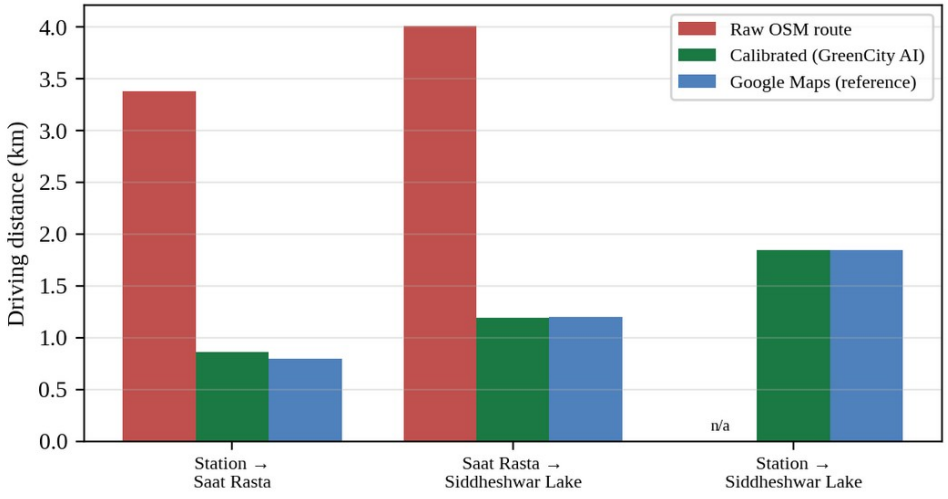


Fig. 4.4: Driving distance before and after detour calibration

E5: Software validation. The back-end automated suite of 69 unit and integration tests passes (69 passed, 0 failed), covering the services and endpoints described in Chapter 3.

4.3 Environmental and Safety Considerations

Because the platform gives advice about outdoor exposure, safety rules are conservative: any severe pollution or extreme heat reduces the score irrespective of other factors, every recommendation carries a provenance record (data source, freshness and confidence) and a medical disclaimer stating that the output is informational and not clinical advice, and sensitive groups are advised to consult a physician. Data sources are model-based estimates and may differ from a nearby ground monitor, so confidence is reported openly. Voice and location data are used only to serve the request, JWT and role-based access protect municipal functions, and administrator actions are audit-logged. The system itself is software running on shared infrastructure, so its direct environmental footprint is small, while its outputs support SDG 11 (healthy access to green space) and SDG 13 (heat and pollution adaptation).

5. TECHNICAL SPECIFICATION, SYNOPSIS & FUTURE PLAN

5.1 Summary of Work Done in Phase-I

During Phase-I the team identified the problem, reviewed related work, designed the three-tier architecture and implemented a working prototype of GreenCity AI. The complete user-facing front-end has been developed and includes the best-time-to-visit engine with a date-of-choice calendar and a 14-day horizon; the CPCB-based environmental dashboard with 72-hour trends and Diwali clean-air watch; calibrated GPS routing with turn-by-turn navigation and automatic recalculation from the live GPS position; nearby-place discovery with a four-park comparison matrix; a multilingual (English, Hindi, Marathi) voice assistant with civic report drafting; and municipal modules for reporting, tracking, analytics, work orders and administration. The user interface is fully localised through three translation files. These modules are supported by a working prototype of the FastAPI back-end and PostgreSQL database. Preliminary simulations and 69 passing automated tests support the correctness of the core logic.

5.2 Technical Specifications, Milestones Achieved and Pending Tasks

Table 5.1: Technical specifications

Parameter	Specification
Study area	Solapur Municipal Corporation, Wards 1–5
Back-end	FastAPI, Python 3.13; REST over HTTP; JWT authentication; RBAC with four roles
Database	PostgreSQL; SQLAlchemy 2.0 ORM; 17 tables
Front-end	Next.js 15 (App Router), React 19, TypeScript; Leaflet maps; Recharts
Forecast horizon	Up to 14 days; hourly resolution; Asia/Kolkata time zone
Air-quality standard	Indian CPCB NAQI from PM2.5 and PM10
Suitability score	0–100; caps: AQI > 200 → ≤ 35, AQI > 400 → 0, T ≥ 37 °C → ≤ 20
Routing	OSM-based; circuity thresholds 1.55 (driving) and 1.35 (walking); calibration factors 1.28 and 1.20
Search radius	1, 2, 3 or 5 km; compare up to 4 places
Languages	English, Hindi, Marathi (UI and voice)
Testing	69 automated pytest tests, all passing

Table 5.2: Milestones achieved and pending tasks

Milestone	Status	Remarks
Problem identification and literature review	Achieved	Chapters 1–2
System architecture and database design	Achieved	17 tables, three-tier design
Best-time engine with safety overrides and calendar	Achieved	14-day horizon
Environmental dashboard (NAQI, 72 h trend)	Achieved	Open-Meteo and CAMS
Calibrated routing and navigation panel	Achieved	Matches reference distances
Multilingual voice assistant and localisation	Achieved	English, Hindi, Marathi
Complete front-end (all pages and role-based dashboards)	Achieved	Municipal, citizen and admin interfaces
Back-end data ingestion, caching and database optimisation	Pending	Phase-II
Validation against public CPCB station records	Pending	MAE / RMSE and confidence levels
Score-weight tuning and routing calibration extension	Pending	Phase-II, all five wards
Assistant evaluation and expanded automated testing	Pending	Phase-II
API security hardening and rate limiting	Pending	Phase-II
Containerised cloud deployment, monitoring and load testing	Pending	Phase-II
Usability study with citizens and officers	Pending	Survey and feedback

5.3 Conclusion and Future Work for Phase-II

Phase-I demonstrates that open weather and atmospheric data, combined with a transparent rule-based score, calibrated routing and a grounded multilingual assistant, can provide Solapur residents with practical, health-protective guidance for using green spaces while giving SMC an integrated management tool. The simulations confirm that the safety overrides behave as specified and that route calibration removes the large detours present in raw OpenStreetMap routes.

Since the front-end has been completed in Phase-I, Phase-II will concentrate on the **back-end**: turning the working prototype services into a fast, reliable, secure and deployable server-side system, using only open data sources and open-source software. The planned work is as follows.

(i) **Data ingestion and caching layer.** Scheduled background jobs will fetch hourly Open-Meteo and CAMS data for all parks and store it in PostgreSQL, so that user requests are served from stored data rather than by calling external services every time. A cache with expiry times, retry with back-off and fallback to the last valid record will reduce response time and keep the platform usable when an external service is slow or unavailable.

(ii) **Database optimisation.** Indexes will be added for frequently used queries, schema changes will be managed through versioned migrations, PostGIS will be evaluated for radius and nearest-park searches, and connection pooling and automated backups will be configured.

(iii) **Data validation.** Model-based PM2.5, PM10 and temperature values will be compared with publicly available CPCB monitoring-station records for Solapur and nearby stations. The mean absolute error (MAE) and root-mean-square error (RMSE) will be reported, and

the confidence levels attached to every recommendation will be set from these results; a bias-correction factor will be applied only if the data justify it.

(iv) **Scoring-engine refinement.** Weights and thresholds will be moved into versioned configuration so that every change is recorded and traceable. The weights will be tuned using historical data and survey feedback, while the safety overrides will remain fixed.

(v) **Routing service improvement.** The circuity-ratio calibration will be tested on more location pairs across all five wards, including walking routes. A locally hosted routing engine built from a Solapur OpenStreetMap extract will be evaluated, and frequently requested routes will be cached.

(vi) **AI assistant back-end evaluation.** A test set of English, Hindi and Marathi queries will be prepared to measure intent detection, tool selection and grounded responses. Every tool call will be logged, and timeouts and safe fallback replies will be added so that no factual answer is produced without a data source.

(vii) **Alert and notification service.** A back-end service with scheduled jobs and new API endpoints will check AQI and heat thresholds for saved parks and prepare alerts and a daily best-time summary for citizens.

(viii) **Security hardening.** Token expiry and refresh, strong password hashing, strict input validation, rate limiting, a restricted cross-origin policy and complete audit logging of administrator and officer actions will be implemented, and the API will be reviewed against commonly known API security risks.

(ix) **API quality and testing.** Error responses and pagination will be standardised, interactive API documentation will be generated, and the automated pytest suite (currently 69 tests) will be extended with more integration tests, a code-coverage report and an automated test run on every code change.

(x) **Deployment and monitoring.** The back-end and database will be packaged in containers and deployed on a cloud platform with HTTPS, health-check endpoints, logging and monitoring. Load testing will be carried out to find the capacity of the system and to remove bottlenecks.

(xi) **Documentation and handover.** An API reference, an administrator manual and a deployment guide will be prepared, and the platform will be demonstrated to SMC officers to collect feedback on adoption.

Table 5.3: Phase-II back-end targets (planned)

Aspect	Phase-I status	Phase-II target
Environmental data access	Fetched from external APIs on request	Served from stored hourly data with cache and fallback
Data accuracy	Model-based values, confidence reported	MAE and RMSE reported against public CPCB records
Automated testing	69 tests passing	Extended suite with coverage report and automated runs
Assistant evaluation	Functional testing only	Multilingual query test set with accuracy measures
Security	JWT, RBAC and audit logs	Rate limiting, token refresh, input validation review
Deployment	Local prototype	Containerised cloud deployment with monitoring and load test

Together, these activities will move GreenCity AI from a working prototype with a complete front-end to a validated, secure and deployable system for Solapur, supported by a robust back-end.

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APPENDIX-I: SUPPORTING MATERIAL

A.1 API Endpoints

Method	Endpoint	Purpose
GET	/api/v1/environment/park-visit-times?park_id, lat, lng, date, force_refresh	Best time to visit a park: score, window, morning vs evening, avoid window, activity ratings, 24-h series, provenance
GET	/api/v1/routing/directions?start_lat, start_lng, dest_lat, dest_lng, mode	Calibrated road distance, duration, polyline and turn-by-turn steps

A.2 Main Web Routes

Route	Purpose
/explore?tab=best-time	Best time to visit with date picker and forecasting
/explore?tab=discovery	Nearby places (1–5 km) and comparison
/map	Live map, GPS tracking and navigation
/environment	Live NAQI, PM2.5/PM10, 72-hour trend, Diwali monitor
/assistant	Multilingual voice and text assistant
/report, /reports	Citizen reporting and issue tracking
/analytics, /priority	Ward analytics and priority action queue
/municipal/dashboard, /municipal/maintenance	Officer KPIs and work orders
/admin, /login	Administration console and authentication

A.3 User Roles

The prototype defines four roles: System Administrator, Municipal Officer, Urban Planner and Citizen, each with a separate landing interface and permissions enforced through role-based access control.

A.4 Test Summary

Command: `PYTHONPATH=. pytest apps/api/tests`. Result: 69 passed, 0 failed.